Web Programming Week 07

Flexbox Principles & Layout Refactoring

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[S1] Building the Block

The Evolution of HTML & CSS Layouts

The Problem with Default Flow

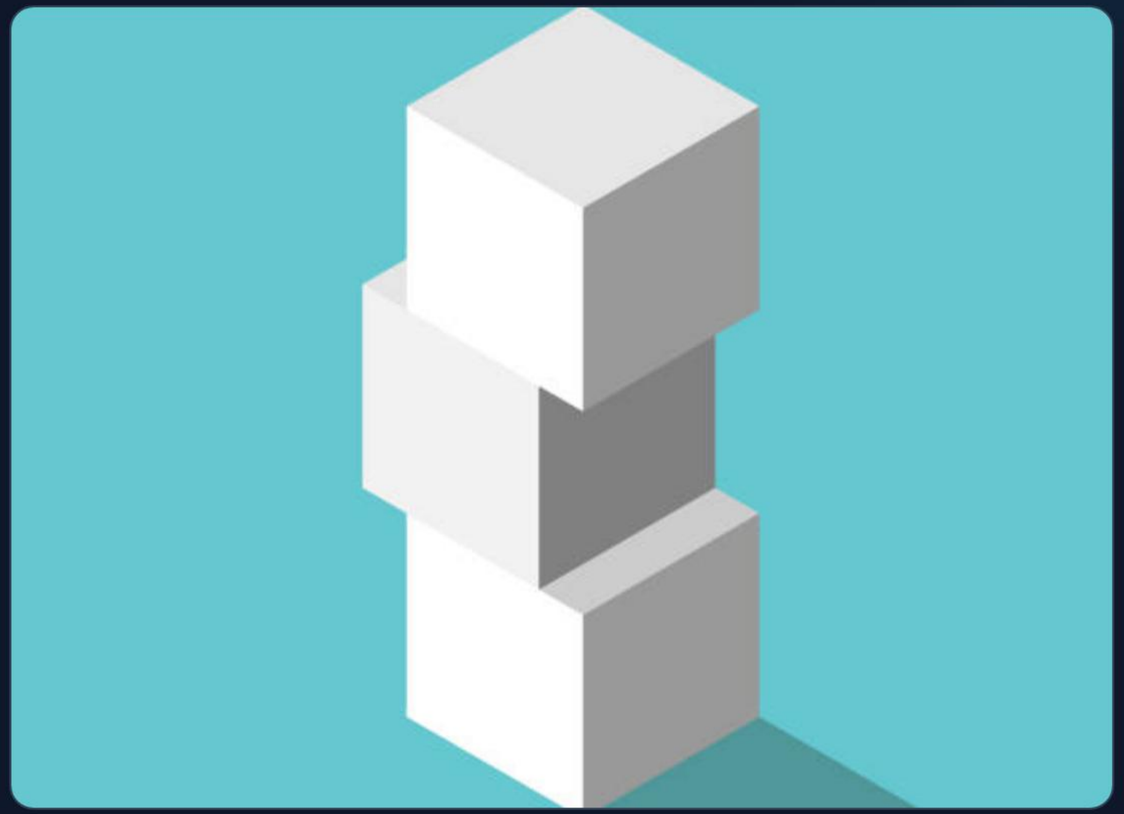
For the past 6 weeks, we have focused on:

→ **Skeleton:** Semantic HTML structure.

→ **Paint:** Applying styles with CSS.

→ **Box Model:** Managing margins and padding.

The Issue: Normal CSS flow stacks block elements vertically. Placing boxes side-by-side often causes unintended line breaks.



The Magic Spell: display: flex

Alignment Manager

Flexbox is a powerful layout model that allows you to position elements freely within a container.

Pixel Perfect

Achieve horizontal rows, vertical columns, or perfectly spaced configurations without complex manual calculations.

Core Flexbox Properties

flex-direction

Determines the primary axis: **row** (horizontal) or **column** (vertical).

justify-content

Distributes space along the **Main Axis** (Center, Space-between, Space-around).

align-items

Controls alignment on the **Cross Axis** (Top, Center, or Text Baseline).

The Golden Rule of Flexbox

#1

The Mother Rule

Parent-Child Relationship

Flexbox operates strictly on hierarchy. Always declare **display: flex;** on the **Parent Container**.

DON'T apply it to the child itself to align the child!

[S2] Flexbox Lab

Live Demo & Responsive Foundations

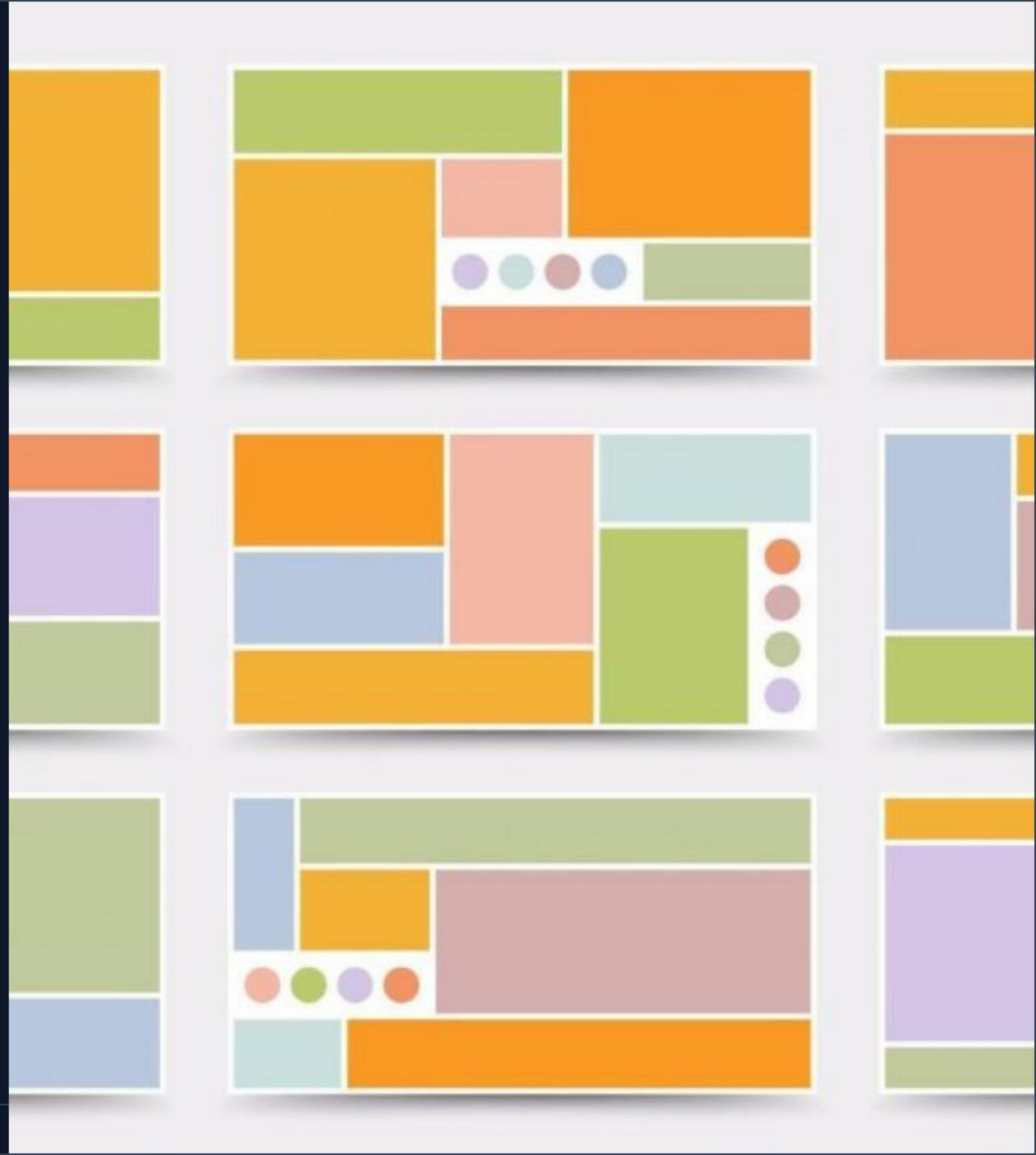
Managing Layout Wrap

The **flex-wrap** property is critical for modern design:

→ **nowrap:** Squashes items to fit a single line.

→ **wrap:** Moves items to a new line when space ends.

Foundation of Responsive Web Design.



[S3] Layout Refactoring

Fixing Spaghetti Code with DevTools

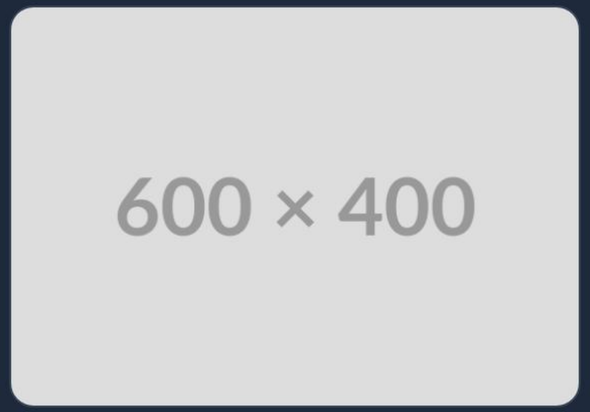
The Developer Tools Workflow

Inspect (F12)

Open the diagnostic panel to see real-time HTML/CSS.

Tweak & Verify

Modify styles in-browser to see instant fixes before coding.



Commit Changes

Copy working styles from DevTools back to your .css file.

Diagnosing & Fixing Overflow

Condition	Description	Solution
The Bug	Image piercing parent container.	Remove fixed pixel widths.
The Cause	Rigid `width: 1000px;`	Ignore parent's flexibility.
The Fix	`.profile-img { max-width: 100%; height: auto; }`	

Questions?

Mastering Flexbox & Clean Refactoring

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Image Sources

https://media.istockphoto.com/id/1194635344/vector/three-white-cubes-stacked.jpg?s=612×612&w=0&k=20&c=edlkv5uJCZvUD2crOSu45zv9-bHat9pBk6Hv8nYTwR0=

Source: www.istockphoto.com

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